What is Word2Vec?

Word2Vec is a technique that learns **word embeddings** (dense numerical vectors) from text.
Instead of manually creating vectors, Word2Vec **learns them automatically by looking at the context in which words appear.**

Simple Definition

Word2Vec is a neural network-based algorithm that converts words into dense vectors such that words appearing in similar contexts have similar vector representations.

Core Idea

Consider these sentences:

```
The cat drinks milk
The dog drinks milk
The cat eats fish
The dog eats meat
```

Word2Vec notices that:

- **cat** and **dog** appear in similar contexts
-
- Both occur near words like "drinks", "milk", "eats"
-

So it learns vectors like:

```
cat = [0.82, -0.15, 0.67]
dog = [0.79, -0.12, 0.70]
```

Since their contexts are similar, their vectors become similar.

Intuition

1) CBOW
2) Skip

Think of a child learning language.

The child may never see a dictionary definition of "cat" or "dog".

Instead, they observe:

```
The cat drinks milk.
Dog drinks milk.

Cat runs.
Dog runs.

Cat eats.
Dog eats.
```

The child realizes:

```
Cat ≈ Dog
```

Word2Vec learns in the same way.

How Does Word2Vec Learn?

It uses a small neural network and learns from surrounding words.

There are two main approaches:

1. CBOW (Continuous Bag of Words)

Predict the center word from surrounding words.

Example:

```
The ___ drinks milk
```

Input:

```
The, drinks, milk
```

Output:

```
cat
```

CBOW learns:

```
Context → Target Word
```

2. Skip-Gram

Predict surrounding words from the center word.

Sentence:

```
The cat drinks milk
```

Input:

```
cat
```

Outputs:

```
The
drinks
```

Skip-Gram learns:

```
Target Word → Context
```

Semantic meaning

Famous Word2Vec Example

After training on a large corpus:

```
King  - Man + Woman ≈ Queen
```

Similarly:

```
Paris  - France + Italy ≈ Rome
```

These relationships emerge automatically from the learned embeddings.

Word2Vec vs TF-IDF

Feature	TF-IDF	Word2Vec
Captures word importance	✅	❌

Captures semantic meaning	❌	✅
Dense vectors	❌	✅
Sparse vectors	✅	❌
Learns context	❌	✅
Used in Deep Learning	Limited	Very Common